PHARMACEUTICAL ANALYSIS

UNIT 3 NOTES

PRECIPITATION TITRATION

- MOHR'S METHOD
- VOLHARD'S METHOD
- FAZAN'S METHOD

COMPLEX METRIC TITRATION

- METAL ION INDICATORS
- MASKING AND DEMASKING AGENTS

GRAVIMETRY

- PRINCIPLE OF GRAVIMETRY
- COPRECIPITATION & POST PRECIPITATION
- DIAZOTIZATION TITRATION

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PRECIPITATION TITRATION

- The word precipitation refers to formation of solid mass in a liquid.
- The solid formed is termed as Precipitate.
- Precipitation titration is a type of titration which involves the formation of precipitate during titration.
- Since, precipitation titration is mainly used $AgNO_3$ (Silver Nitrate) as titrant, hence it is also called 'Argentometric Titration'.

$$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$$

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Conditions For Precipitation Titration

- Precipitate must be insoluble
- Precipitation reaction must be rapid
- It must be possible to detect the end point.

Types / Methods of Precipitation Titration

- Mohr's Method
- Volhard's and Modified Volhard's Method
- Fajan's Method

ARGENTOMETRIC TITRATION

- Titration which involves silver as titrant are known as Argentometric Titration.
- Argentometry is derived from a greek word 'Argentum' means 'Silver'
- This is why silver is denoted by 'Ag' in periodic table.

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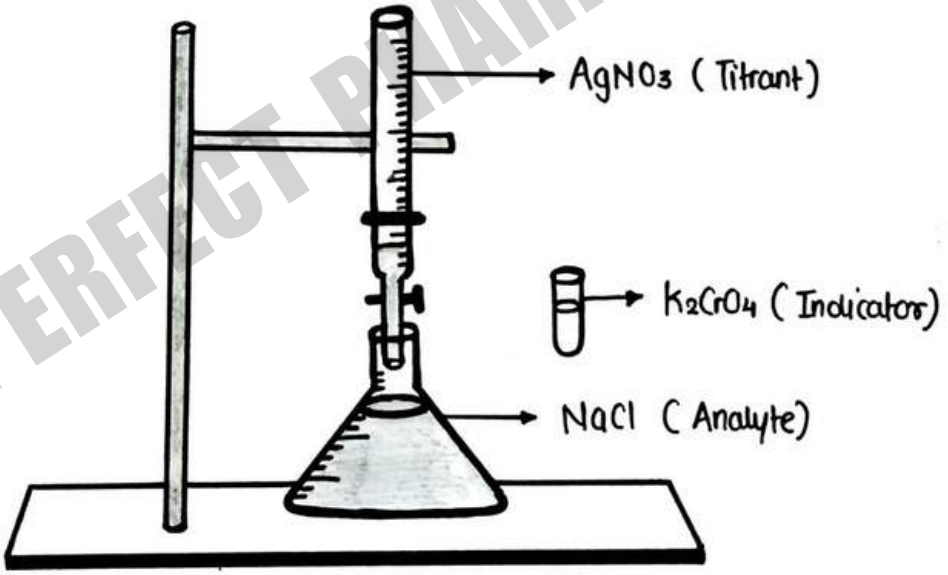
MOHR'S METHOD

- The method was given by Sir karl Friedrich Mohr.
- A precipitation titration in which silver nitrate is used as titrant & chromate ions as indicator is called as Mohr's Method.
- The method is mainly used for the determination of chlorides and bromides (Cl and Br).

Titrant : $AgNO_3$

Analyte : KCl, NaCl, KBr, NaBr

Indicator : K_2CrO_4 or Na_2CrO_4



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Methodology / Procedure

- Let's suppose we are titrating 0.1 M NaCl using 0.1 M $AgNO_3$ as titrant and K_2CrO_4 as indicator.
- As the titration starts, Silver Nitrate solution is slowly added to Sodium Chloride and a precipitate of Silver Chloride forms.

$$AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$$

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- Now when all the chloride ions gets precipitated, then additional silver ion react with chromate ions (indicator) and form a reddish brown precipitate of silver chromate and this gives our end point.

$$2\,AgNO_3 + K_2CrO_4 \longrightarrow Ag_2CrO_4 + 2KNO_3$$

Conditions For Mohr's Method

- In Mohr's Method titration must be carried out in a neutral medium neither acidic nor basic.

Because ① In basic medium $Ag^+ + OH^- \rightarrow AgOH$

② In acidic medium, $CrO_4^{2-} + H^+ \rightarrow H_2CrO_4$

NOTE

- The K_{sp} of AgCl = 1.8×10^{-10}
- The K_{sp} of Ag_2CrO_4 = 1.2×10^{-12}

Although K_{sp} value of Ag_2CrO_4 is lower than K_{sp} value of AgCl and it should be precipitated first, but since the concentration of Cl^- ion in the analyte solution is much more higher compare to CrO_4^{2-}, hence Ag^+ react with Cl^- ions first & that's why AgCl formed before Ag_2CrO_4

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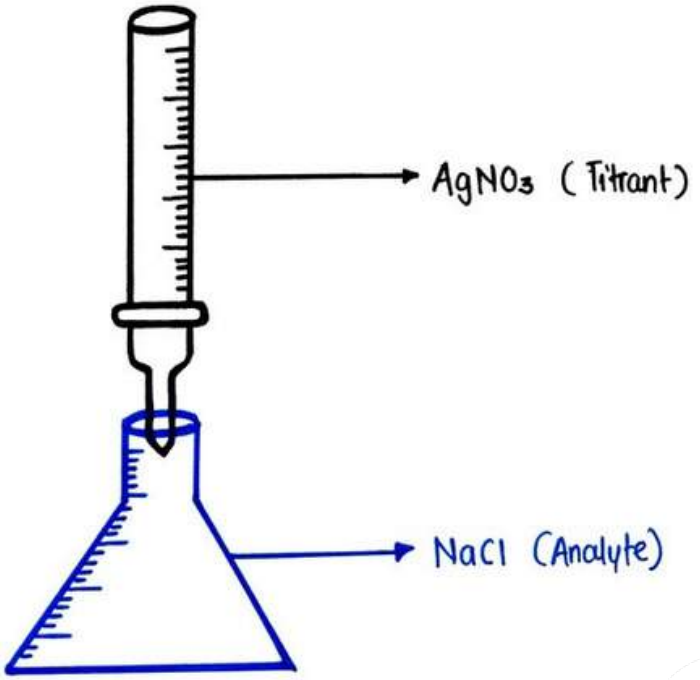
VOLHARD METHOD

- The method was given by german scientist Jacob Volhard
- This is an indirect method of titration involves back titration.
- Volhard method is mainly used in the determination of Halides [Cl, Br, I)
- Precipitation titration by volhard method completed in two steps :

STEP - I

In first step excess amount of $AgNO_3$ is used for the titration with NaCl solution.

$$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3 + AgNO_3 \text{ (excess)}$$



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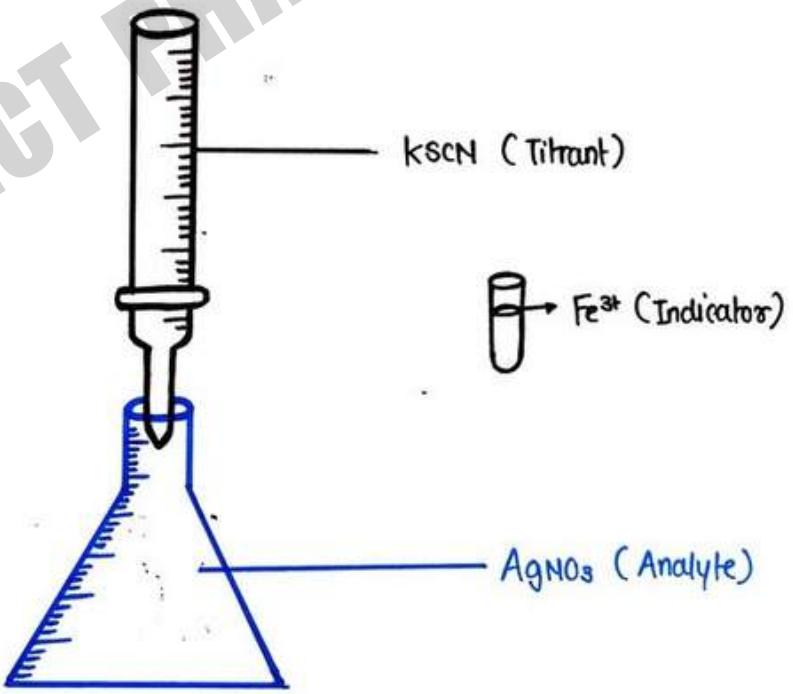
STEP- II

- Now the excess amount of $AgNO_3$ which gets remained in step-1 is titrated using KSCN as titrant and Fe^{3+} as indicator.

$$AgNO_3 + KSCN \longrightarrow AgSCN + KNO_3$$

- Now when all $AgNO_3$ is consumed by KSCN (potassium thiocynate) then extra thiocynate ions react with Fe^{3+} and form ferrous thiocyanate ($Fe(SCN)^{2+}$) which is a red complex and this gives our end point.

$$Fe^{3+} + SCN^- \longrightarrow Fe(SCN)^{2+}$$



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Conditions for Volhard's Method

The titration must be performed in a acidic medium, so that complex formed [$Fe(SCN)^{2+}$] will be stable.

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MODIFIED VOLHARD'S METHOD

- In modified volhard's method we add some wetting agents like chloroform, nitrobenzene etc.
- Since, AgCl is also present in the analyte and it may solubilise which affect our end point, and addition of these wetting agents prevents the solubilization of AgCl.

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FAJAN'S METHOD

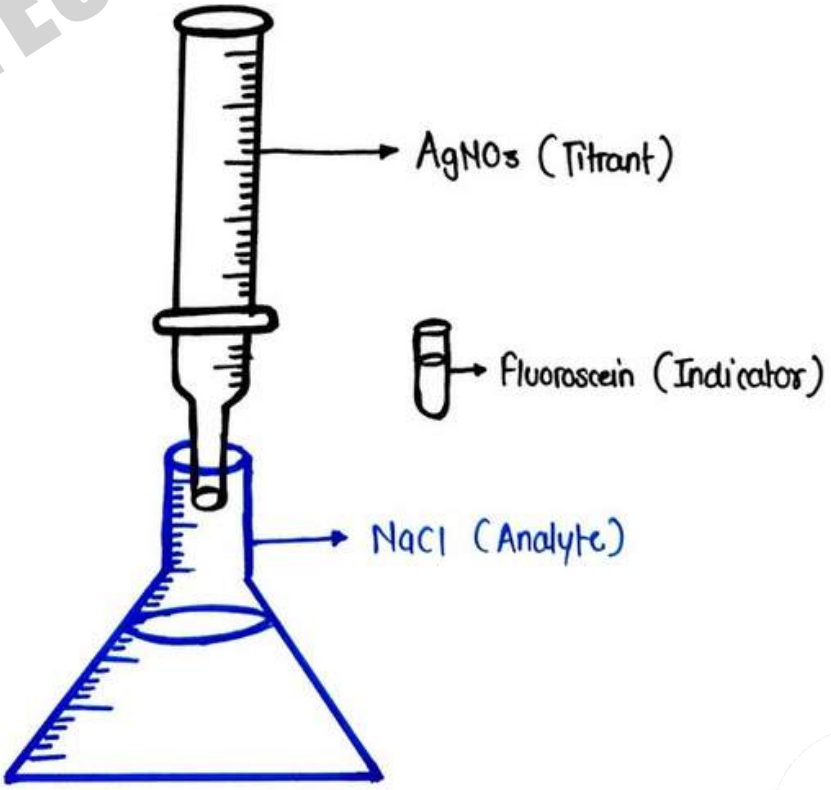
- The method was given by Sir kazimierz Fajan.
- The method is based upon the theory of adsorbtion indicators.
- The method is used for the determination of halides (Cl, Br, I) using $AgNO_3$ as titrant and Fluorescein as indicator.
- At the end point, adsorbtion indicators gets adsorb on the surface of AgCl and changes the colour of precipitate.

Titrant : $AgNO_3$

Analyte : NaCl

Indicator : Fluorescein

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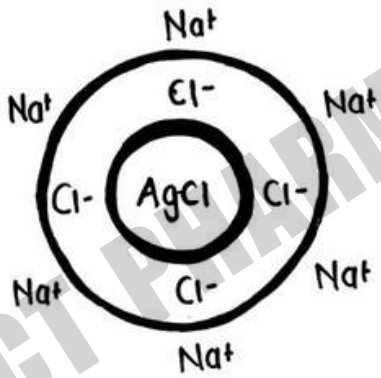
Procedure

When all the NaCl solution in the analyte is consumed by $AgNO_3$ and converted into AgCl, then fluoroscein indicator gets adsorb on the surface of AgCl & changes the colour to pink.

① Before Endpoint

$AgNO_3$ reacts with ~~Agcl~~ an NaCl & converted into AgCl.

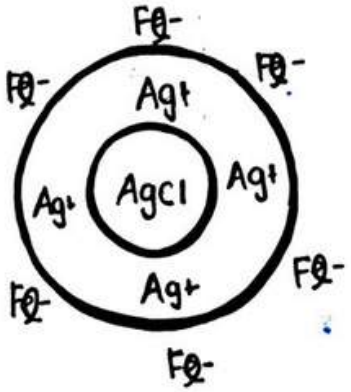
$$AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$$



② At Endpoint

Fluoroscein indicator adsorbs on the surface of AgCl & change colour.

$$Ag^+ + Fl^- \longrightarrow AgFl$$



Fl^- = Fluoroscein

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COMPLEXOMETRIC TITRATION

- Complexometric titration is a type of titration which is based on the complex formation between the analyte and titrant.
- It is also known as Chelatometric Titration.
- Complexometric Titration are mainly used for the determination of metal ions in a solution.
- End point of the titration is determined by change in colour of the solution due to complexometric reaction.
- Since EDTA (Ethylene diamene tetra acetic acid) is mainly used as titrant in complexometric titration, hence it is also called EDTA Titration.

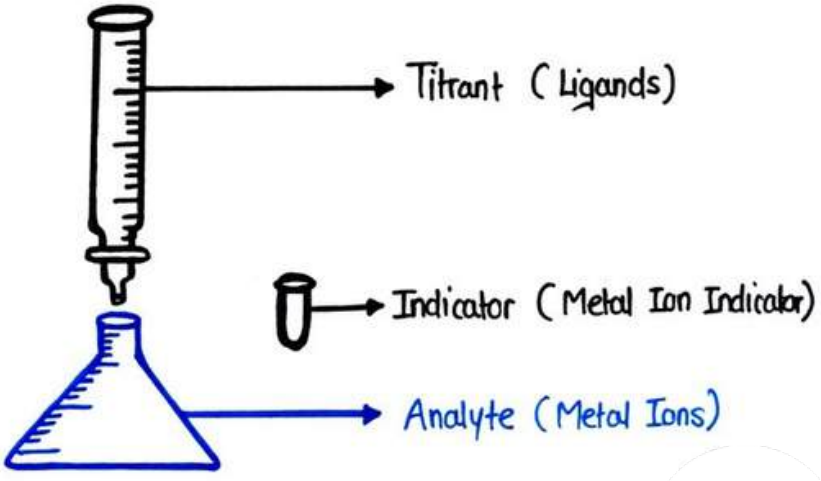
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In Complexometric Titration we can simply say :

Analyte : Metal Ions

Titrant : Ligands or Complexing Agents (Mainly EDTA)

Indicator : Metal Ion Indicators



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Types of Ligands in Complexometric Titration

Ligands are classified on the basis of number of donor groups :

① Unidentate Ligands / Monodentate Ligands
② Bidentate Ligands
③ Multidentate / Polydentate Ligands

Monodentate Ligands

- It is also known as unidentate Ligands.
- When the ligands has only one donor atom, it is called monodentate.
- examples : NH_3, H_2O etc.

Bidentate Ligands

- When the ligands has two donor atoms, it is called Bidentate ligands
- example : Ethylene Diamene ($H_2N:-CH_2-CH_2-:NH_2$)

Multidentate Ligands

- It is also known as Polydentate ligands.
- When the ligands have more than 2 donor atoms, it is called as Multidentate ligands.
- example EDTA (Ethylene Diamene Tetraacetic Acid)

- ~~Mono~~ multidentate ligands are also known as Chelates

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Types of Ligands in Complexometric Titration

Ligands are classified on the basis of number of donor groups :

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Monodentate Ligands

- It is also known as unidentate Ligands.
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- Examples : NH_3, H_2O etc.

Bidentate Ligands

- When the ligands has two donor atoms, it is called Bidentate ligands.
- example : Ethylene Diamene ($H_2N:-CH_2-CH_2-:NH_2$)

Multidentate Ligands

- It is also known as Polydentate ligands.
- When the ligands have more than 2 donor atoms, it is called as Multidentate ligands.
- example EDTA (Ethylene Diamene Tetraacetic Acid)

- Multi~~Mono~~dentate ligands are also known as Chelates

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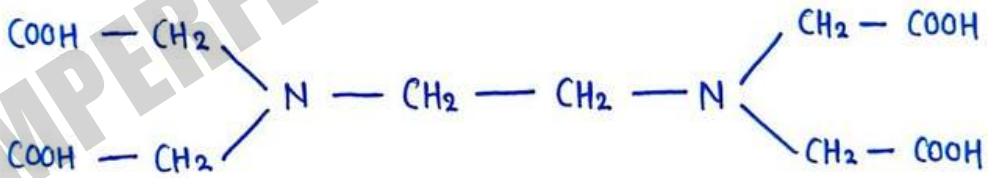
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EDTA

- EDTA stands for Ethylene Diamene Tetraacetic Acid.
- It is a multidentate ligand or we can also call them Chelates.
- Generally it is insoluble in water, hence in complexometric titration we use its salt Disodium EDTA. which is water soluble.
- It is most widely used as titrant in complexometric titrations.

Chemical Formula : $C_{10}H_{16}N_2O_8$
Molecular Weight : 292.24 g/mol

Structure



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Classification of Complexometric Titration

① Direct Titration
② Back Titration / Indirect Titration
③ Replacement Titration
④ Alkalimetric Titration

Direct Titration

- It is the simplest method of complexometric titration.
- In this EDTA is mainly used as titrant.
- In this end point is simply determined by colour change due to indicator.
- It is similar to acid- base titration.
- It is not suitable for slow complexometric titrations.

Back Titration

- It is also known as Indirect titration.
- In this titration is performed by taking excess amount of EDTA
- In this first we add excess amount of EDTA in the analyte, then remaining amount of EDTA is again titrated by solution of second metal ion.

Replacement Titration

- This method is used when direct & back titration don't give sharp end point.
- In this titration, metal present in the analyte displaces another metal from metal- EDTA complex.

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Alkalimetric Titration

- It is used for the determination of anions which generally do not react with EDTA chelate.
- In this method, the free H^+ ions get liberated during titration.
- The free H^+ ions is then titrated with standard solution of alkali using suitable indicator.

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Metal Ion Indicators

- Indicators that are used in complexometric titration are known as Metal Ion Indicators.
- It is also known as pM Indicators

Mechanism of Action of pM Indicators

- Like all other indicators, pM indicators also shows colour change at the endpoint, but its working is somehow different :-

- First we add indicator in the analyte (metal ions) solution, now indicator formed a complex with analyte and colour of this complex is wine red, hence we can say before end point the colour of the analyte is wine red.

- **At Endpoint** : At the Endpoint, the titrant (ligands) breaks the bond between indicator and analyte and titrant self forms a complex with analyte which is called Metal Ion Complex or Ligand- metal ion complex.

- Now since indicator gets free, and the free form of indicator have blue colour, hence we can say at the endpoint, indicator shows blue colour.

- Some commonly used indicators in complexometric titrations are : Eriochrome black T, Calmagite, Murexide, Catechol Violet etc.

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MASKING & DEMASKING AGENTS

- Masking and demasking agents are generally used when analyte solution contains two or more than two metal ions.
- Let consider we have a solution containing three different metal ions A, B and C, now if we directly titrate the analyte, then we will get the concentration of all three metals A, B, C
- But in the case, if we have to determine the concentration of A, B and C individually then we have to use Masking and Demasking Agents.

Masking Agents

- Masking agents are the reagents that forms a complex with some components of the analyte and protect them from reaction with EDTA
- Masking agents are mainly used when the analyte solution contains two or more metal ions.

Demasking Agents

- Demasking agents are the reagents that releases metal ions from masking agents.
- They break the complex formed between metal ion & masking agent.

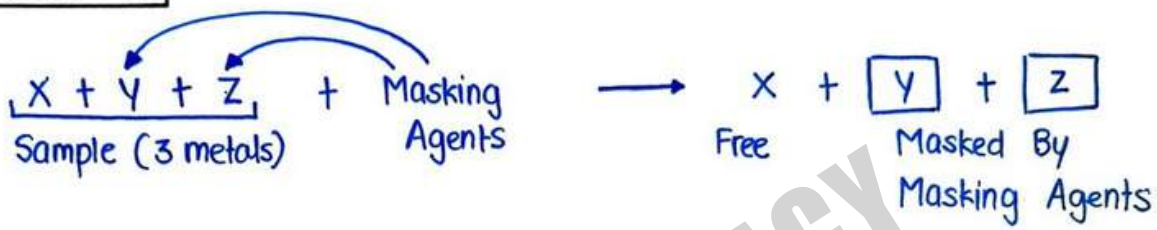
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Mechanism of Masking and Demasking Agents

Let's consider we have an analyte solution containing three metal ions X, Y and Z, and we have to find their concentration individually.

STEP - I



STEP - II



STEP - III



STEP - IV



Now Z is also analyzed same as X and Y

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Estimation of Magnesium Sulphate

Estimation of magnesium sulphate is done by complexometric titration using EDTA as titrant.

Chemical Formula : $MgSO_4 \cdot 7H_2O$
Molecular weight : 246.6 g/mol

Properties of Magnesium Sulphate

- It is colourless or white crystalline powder.
- It is hygroscopic in nature
- It is odourless
- It has bitter taste

Principle

$$Mg^{2+} + EDTA \longrightarrow Mg\text{-}EDTA$$

Titrant : Disodium EDTA

Analyte : Magnesium Sulphate

Indicator : Mordant black II

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Procedure

- Weigh accurately about 0·3 g of magnesium sulphate
- Dissolve in 50 ml of water
- Add 10 ml of strong ammonia - ammonium chloride solution.
- Titrate the solution with 0·05 M disodium edetate using 0·1 g of mordant black II as indicator until a blue colour is obtained.

Uses

- It is used as cathartics
- It can also be used as beauty products

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Estimation Of Calcium Gluconate

Estimation of calcium gluconate is done by complexometric titration using EDTA as titrant.

Chemical Formula : $C_{12}H_{22}CaO_{14}$
Molecular Weight : 430.373 g/mol

Properties of Calcium Gluconate

- It appears as white crystalline powder.
- It is odourless
- It is tasteless
- It is soluble in water
- It is insoluble in ~~water~~

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Principle

$$Ca^{2+} + EDTA \longrightarrow Ca\text{-}EDTA$$

Titrant : Disodium EDTA

Analyte : Calcium Gluconate

Indicator : Mordant Black II or Solochrome Black T indicator

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Procedure

- Weigh accurately about 0.5 g of calcium gluconate in a conical flask and dissolve in 50 ml of warm water.
- Add 5 ml of 0.05 M magnesium sulphate.
- Add 10 ml strong ammonia - ammonium chloride solution.
- Add 2-3 drops Mordant black II as indicator.
- Titrate the solution with disodium EDTA until colour changes to blue.

Uses

- Used in case of calcium deficiency
- Used in bone development.

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GRAVIMETRIC ANALYSIS

- The word gravimetry refers to mass.
- Gravimetric analysis is method of analysis in which we determine the quantity of analyte by measurment of mass.
- In gravimetric analysis analyte is converted into precipitate.

Methods of Gravimetric Analysis

① Precipitation Gravimetry
② Volatilization Gravimetry
③ Electrogravimetry

Advantages of Gravimetric Analysis

- Gravimetric analysis is more accurate and more precise compared to volumetric analysis.
- It is an absolute method which involves direct measurment.
- Less chances of mistake.
- Generally not affected by temperature.

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Steps Involved In Gravimetric Analysis

The estimation of gravimetric analysis consist a number of steps to get accurate results, it includes :

- Sampling
- Preparation of solution
- Precipitation
- Digestion / Ostwald Ripening
- Filteration
- Washing
- Drying or Ignition
- Weighing
- Calculations

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Sampling

- The sample which is weighed for gravimetric analysis should be very small.
- The sample should be homogenous.
- The sample must be in powder form.
- It should not be very expensive.

Preparation of Solution

- To dissolve the sample, take a clean beaker and transferred the weighed sample completely into beaker.
- Add sufficient water to get a clear sample.
- temperature, pH, pressure should be maintained at normal.
- If necessary, then heat the solution.

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Precipitation

- In this process, we generally add a precipitating agent to the sample solution.
- The ideal precipitating agent must react with analyte to form precipitate.
- Precipitation generally carried out in a resistance glass beaker.
- The completeness of precipitation is checked by adding a few drops of precipitating agent on the side of beaker wall.
- It is very necessary to check the completeness of precipitation.

Digestion

- Digestion process is also known as Ostwald Ripening.
- During digestion, the precipitate is left hot for 30 min - 1 hour
- Digestion involves dissolution of small particles.
- Digestion process is very helpful in the case of colloidal particles.
- Digestion, sometimes increases the problem of post-precipitation.

Filteration

- In this step, precipitate is separated from the analyte solution.
- Various types of filter media used in this process.
- The choice of filter media is depend upon the nature of precipitate.
- The different filter media used in gravimetric analysis are :

① Filter Paper
② Filter Pulp
③ Filter Mats etc.

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Washing

- After filtering the precipitate, the impurities at the surface of the precipitate can be removed by washing the precipitate.
- One should be very careful not to use too much water during washing, because part of the precipitate may lost.
- Water is not suitable for many precipitate as washing solution.
- In such cases dilute nitric acid, ammonium nitrate etc. can be used as washing solution.

Drying or Ignition

- The main purpose of drying or ignition is to convert the precipitate in a constant composition.
- It is heated to remove the water
- Drying term is used when temperature is below 250°C
- Ignition term is used when temperature is above 250°C and below 1200° C.
- Drying or Ignition is depend upon the nature of precipitate.

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Weighing and Calculation

- After drying the precipitate is then cooled at room temperature and weighed accurately on the analytical balance.
- The calculations are generally made in terms of percentage.
- Calculations of analyte content requires knowledge of :

- Chemistry
- Stoichiometry
- Composition of precipitate

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PURITY OF PRECIPITATE

- When the precipitate is separate out from the solution for further process like digestion, drying and weighing, it is not always in pure form, some impurities are also mixed with it.
- The amount of impurities depends upon the nature of precipitate and condition of precipitation.
- Purity of the precipitate is mainly affected by two sources

① Co- Precipitation
② Post - Precipitation

CO - PRECIPITATION

- Co- Precipitation is a process in which during the precipitation, impurities also gets precipitated along with our main precipitate.
- The problem of co- precipitation usually occurs when more than 2-3 compounds are mixed in our analyte solution.
- The chances of co- precipitation is generally higher during gravimetric analysis

Types/ Mechanism of Co- Precipitation

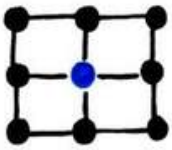
① Inclusion
② Occlusion
③ Surface Adsorbtion
④ Mechanical Entrapment

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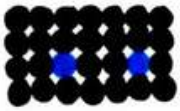
Inclusion

Inclusion is a type of co-precipitation in which impurity occupies the crystal lattice site in the crystal lattice structure of precipitate by replacing an element.



Occlusion

Occlusion is a type of co-precipitation in which the impurities gets trapped between the crystals due to rapid crystal growth.



Surface Adsorbtion

Surface adsorbtion is a type of co-precipitation in which impurities gets adsorbed on the surface of the precipitate.

Mechanical Entrapment

It occurs when crystal growth is very fast, several crystals grow together and trap a portion of impurities.

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POST PRECIPITATION

- The process by which an impurity gets deposited on the surface of the desired precipitate after its formation is called Post - Precipitation.
- It occurs after the precipitation is completed.
- It generally occurs when the precipitate formed is left in the mother liquor for a long time.
- The problem of post-precipitation is increases during digestion process.

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Basic Difference between Co- Precipitation & Post Precipitation

	Co - Precipitation	Post - Precipitation
Definition	It is a type of coprecipitation in which impurity gets precipitated together with main precipitate	In post - precipitation impurity precipitated after the main precipitation completed
Degree of Contamination	High	Low
Time of Precipitation	During the Precipitation	After the Precipitation

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Estimation Of Barium Sulphate

Chemical Formula	$BaSO_4$
Molecular Weight	233.4 g/mol

Properties

- It is a fine, heavy, white colour powder
- It is odourless
- It is tasteless
- It is insoluble in water

Principle

When dilute Sulphuric acid (H_2SO_4) is added to the dilute solution of barium chloride ($BaCl_2$) a white precipitate of barium sulphate ($BaSO_4$) is formed.

$$BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HCl$$

Procedure

- Pipette out 25 ml of given solution of barium chloride ($BaCl_2$) in 500 ml beaker.
- Now add 0.5 ml concentrated H_2SO_4 and 100 ml distilled water
- If required then heat the solution.
- Now to this solution add dilute sulphuric acid (H_2SO_4) dropwise until the precipitate of $BaSO_4$ (barium sulphate) is not formed.
- Allow the precipitate to settle down.

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- Now filter the precipitate.
- Wash the precipitate 3-4 times with hot water.
- Now dry out the precipitate
- If water is not removed then repeat the process of drying / Ignition.
- Now cool the precipitate, measure the weight and calculate the amount of barium.

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DIAZOTISATION TITRATION

- Diazotitration titration is a method used for determination of primary aromatic amine compounds.
- The diazotization titration is nothing but simply the conversion of primary aromatic amine to a diazonium compound.
- The process was first discovered by Peter Griessin in 1853.
- The method is mainly used for determination of dyes.

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Principle of Diazotisation Titration

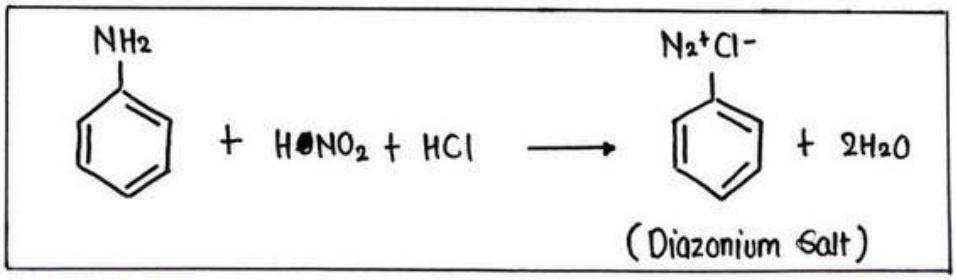
- Primary aromatic amines in the presence of HCl reacts with Sodium Nitrite ($NaNO_2$) and form Diazonium Salt.
- The reaction completed in two steps

STEP - I

$$NaNO_2 + HCl \longrightarrow HNO_2 + NaCl$$

Sodium Nitrite ($NaNO_2$) react with HCl and form Nitrous Acid (HNO_2) and Sodium Chloride ($NaCl$)

STEP - II



The obtained nitrous acid react with primary aromatic amine to form Diazonium Salt.

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End Point Determination in Diazotisation Titration

- In diazotisation titration we use Starch Iodine Paper as indicator.
- At the end point of the titration, starch iodine paper changes the colour into blue.

Starch Iodine Paper

- Starch iodine paper is prepared by dipping the filter paper into Potassium iodide (KI) and starch mucilage solution.
- Following reaction occurs during end point determination

$$KI + HCl \longrightarrow HI + KCl$$

$$2HI + 2HNO_2 \longrightarrow I_2 \uparrow + 2NO + 2H_2O$$

$$I_2 + \text{Starch} \longrightarrow I_2\text{-Starch Complex (Blue)}$$

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Condition For Diazotisation Titration

- Maintainance of temperature is the main condition for diazotisation titration.
- The diazonium salt is not stable at high temperature, so the temperature should be maintained at 0-5° C.

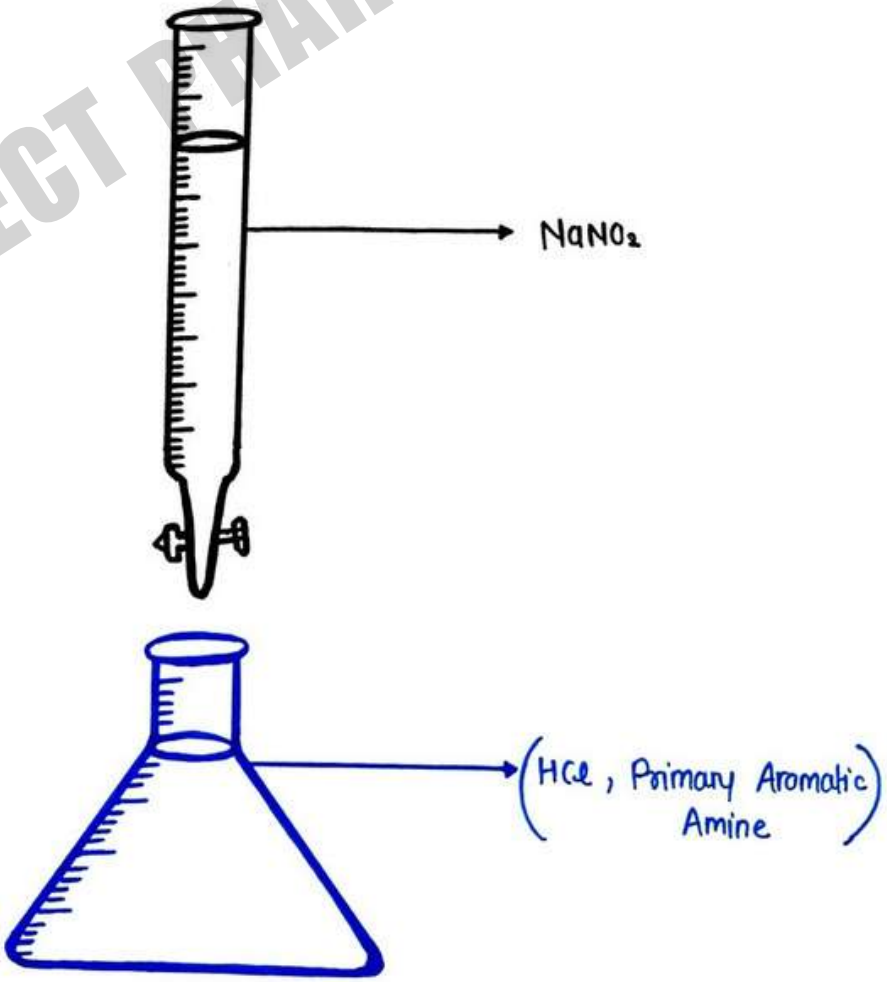
Disadvantages of Diazotisation Titration

- Applicable for very less variety of samples.
- Relatively slow compared to other methods.
- Required to maintain temperature conditions.
- End point detection is very difficult.

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Methodology

- The general procedure is followed by weighing the sample and transferring it into standard flask.
- After that concentrated hydrochloric acid and potassium bromide are added and rest of the volume is filled with distilled water.
- The resulting solution is known as standard solution.
- The appropriate volume of the standard solution is pipette out and temperature is maintained at 0-5°C.
- Now finally the solution is titrated with sodium nitrite solution until starch iodide paper turns into blue colour.



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Applications of Diazotisation Titration

- Used in the determination of sulphonamides
- Used in the determination of chlorophenetamine
- Used in the determination of procaine
- Used in the determination of dopamine
- Used in the determination of ephedrine etc.

NOTE : Since Sodium nitrite is used as Titrant in diazotisation titration, hence it is also known as Nitrite Titration.

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THANK YOU

FOR CHOOSING IMPERFECT PHARMACY AS YOUR STUDY PARTNER

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